

Question ID: 717a1964

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$z^2 + 10z - 24 = 0$$

What is one of the solutions to the given equation?

Correct Answer: 2, -12

Rationale

The correct answer is either **2** or **−12**. The left-hand side of the given equation can be rewritten by factoring. The two values that multiply to **−24** and add to **10** are **12** and **−2**. It follows that the given equation can be rewritten as $(z + 12)(z - 2) = 0$. Setting each factor equal to **0** yields two equations: $z + 12 = 0$ and $z - 2 = 0$. Subtracting **12** from both sides of the equation $z + 12 = 0$ results in $z = -12$. Adding **2** to both sides of the equation $z - 2 = 0$ results in $z = 2$. Note that 2 and -12 are examples of ways to enter a correct answer.